

**U.S. Environmental Protection Agency
Office of Air and Radiation
Office of Radiation and Indoor Air
Indoor Environments Division**

**Development of EPA's Simplified Indoor Air
Chemistry Simulator (SIACS) Model**

Version 1.0

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A. PROJECT MANAGEMENT

A1. Distribution List

Copies of this plan and all revisions will be sent to the following individuals:

Table 1. Distribution list and contact information

NAME	ORGANIZATION	CONTACT INFORMATION
Vito Ilacqua	EPA/OAR/ORIA/IED/CSA	Ilacqua.Vito@epa.gov
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Alisa Smith	EPA/OAR/ORIA/IED	Smith.Alisa@epa.gov

A2. Project/Task Organization

The work described herein is being conducted by U.S. EPA staff in the Office of Radiation and Indoor Air (ORIA), Indoor Environments Division (IED). This Quality Assurance Project Plan (QAPP) was prepared in accordance with EPA *Guidance for Quality Assurance Project Plans for Modeling EPA QA/G-5M* and describes the process for developing a simple model for characterizing indoor air chemistry. Specifically, the QAPP summarizes various publicly available data sources which serve as inputs to this model, quality criteria and considerations for input and output data, and processes for validation and verification of the model and its output. The organizational chart for the SIACS model development is presented in Figure 1 and a description of project personnel's roles and responsibilities are detailed in Table 2.

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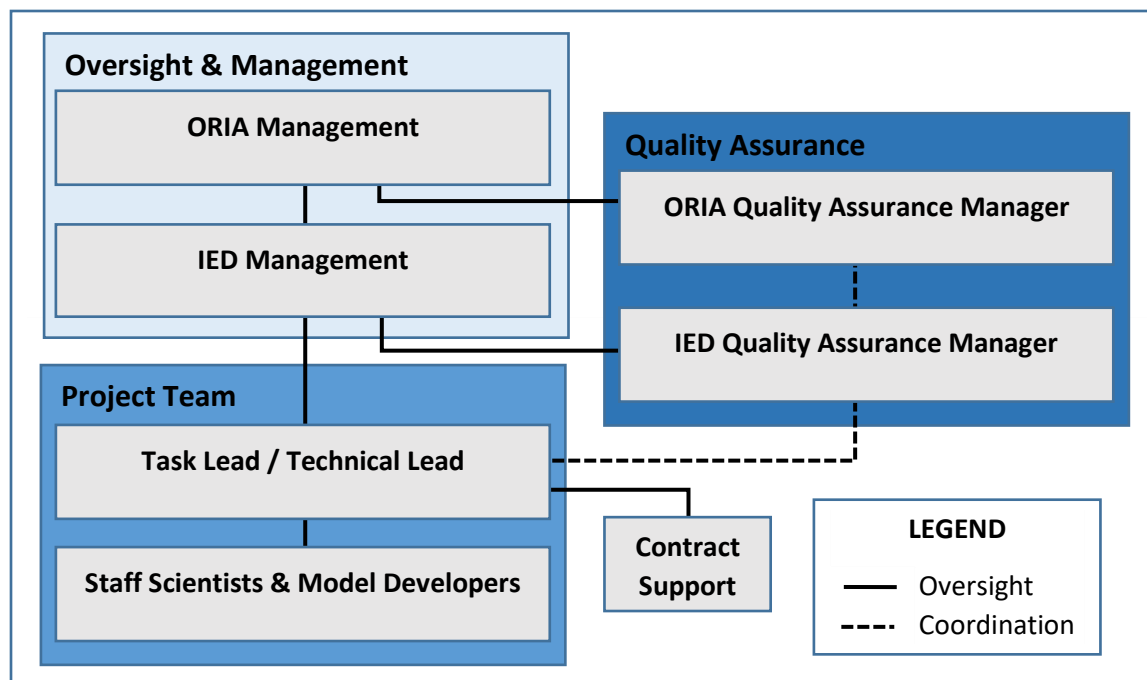


Figure 1. Project organization chart

Table 2. Roles and Responsibilities for SIACS Modeling Effort

Project Personnel	Role	Responsibility
Vito Ilacqua	Principal Investigator & Technical Lead	Set overall project direction; manage assign project outputs (software products and publications); communicate project details to external parties; develop source code; manage software deployment; mentor students and research fellows
Serena Chung	ORD Scientist & Model Developer	Develop source code; manage software deployment
Jordan Zambrana	IED Scientist	Design, develop, test, document and deploy software.
Nicole Scharko	IED Scientist; (AAAS Fellow)	Design, develop, test, document and deploy software.
Daniel Malashock	IED Scientist; QA Lead	Design, develop, test, document and deploy software. Prepares QAPP and revises as necessary.
EPA Contractor(s)	SRC or others as needed	Consultation on source code development and deployment
Alisa Smith	IED Quality Assurance Coordinator (Acting)	Reviews and approves QAPP, performs audits, as appropriate, prepares audit reports, tracks resulting corrective actions, reviews and approves scientific and technical products cleared by the technical lead

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Project Personnel	Role	Responsibility
Jennifer Mosser	ORIA Quality Assurance Manager	Reviews and provides final approval of QAPP, performs audits, as appropriate, prepares audit reports, and tracks resulting corrective actions,

A3. Problem Definition/Background

As people in developed countries spend nearly 90% of their time indoors (EPA, 1989.), exposure to air pollutants largely occurs in the indoor environment. Building characteristics (e.g., air exchange rates, building materials), consumer products (carpets, furniture, cleaning products, paints, houseplants, personal care items), indoor activities (cooking, cleaning, smoking), along with both indoor and ambient air characteristics, all affect indoor air chemistry and thus air pollutant exposure. Understanding how the interaction of these and other critical factors ultimately impact indoor air pollutant concentrations is important. For this reason, a number of indoor air pollutant measurement experiments and models are being developed. In some cases, these are focused on specific, narrow parameters or time scales while others, require extensive technical data for inputs along with expertise for interpretation of results.

The Indoor Environments Division at the U.S. Environmental Protection Agency looks to meet the need for a simple-to-run model as a first screening tool with the Simplified Indoor Air Chemistry Simulator (SIACS). SIACS is a reduced-form model designed to be open source and runs on R statistical software. The model, based on the SAPRC-99 condensed mechanism (Carter 2000), will use about 78 chemical species (some of which are grouped in classes of species with similar reactivity) and 211 photochemical reactions. The model is intended to be a first step in examining conditions that can affect indoor air chemistry and may be used for hypothesis generation and identifying the need for more complex and computationally intensive analyses with other proprietary models.

The first generation SIACS model will be focused on researchers and public health practitioners; however, the intent is to ultimately provide a publicly available, open-source model for developing guidelines and recommendations for indoor air quality. Such recommendations could include, occupant behaviors, consumer products, building materials, or adapting building design and use – such as the risks and benefits of achieving higher energy efficiency through controlling infiltration and exfiltration. Additionally, indoor air quality practitioners may use the model as a tool to explore plausible scenarios that account for unexplained indoor sources.

A4. Project/Task Description

SIACS is a reduced-form model designed to be open source. The current beta version and final open-source model will be operated on R statistical software. The project will also develop a graphic user interface (GUI) for the model to be operated via web application. Accompanying the model will be user guides, technical documentation, and other guidance materials. The

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project will also create a publicly accessible database for input parameters gathered from the literature.

An iterative software development process will be used to develop the SIACS model and accompanying web application. This approach is centered around engagement and feedback from end users at defined increments of progress. These engagements may contribute to refinement of the model based on user-defined needs and/or user-contributed information that may potentially improve modeling approaches. Additionally, these engagements may also identify additional secondary data sources that may be considered for inclusion in the SIACS database. Identified data sources will be evaluated against data quality objectives (See section A7. Quality Objectives for Secondary Data.)

These deployment and feedback approaches are paired with a project management approach that allows for requirements and solutions to evolve over the course of the project and as a result of the collective efforts of the team. The project is organized into three phases as follows, which are each discussed in greater detail throughout this QAPP:

- **Phase 1 – Modeling Needs and Requirements Analysis**
 - Identify customer and modeling needs (completed)
 - Identify project-specific requirements (completed)
 - Complete a QAPP
- **Phase 2 – Model & Database Development**
 - **Model Design**
 - Theoretical development (completed)
 - Mathematical formulation (completed)
 - Identification of needed input data and parameter values
 - **Model Coding**
 - Development of data management and software configuration
 - Model programming
 - Base code & packages
 - Refinement of model
 - Development of additional model components as needed and resources allow (e.g., to account for secondary organic aerosol (SOA) formation and light)
 - Examples for end-users
 - Monte carlo simulation capability
 - Model optimization & performance
 - **Model Testing**
 - Evaluate Uncertainty impact of measurement uncertainty
 - Determine if Model Meets Performance Criteria
 - Assess the capability of the SIACS model to capture variability in time-resolved concentrations of indoor air species

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- Compare and contrast model results to published literature on indoor air chemistry.
- Summarize findings on what covariates played important roles across the sites, response variables, and modeling methods
 - Calibrate the model
- **Database Development**
 - Gather input parameters from the literature
 - Collect new and existing data from stakeholders and partners
- **Stakeholder Engagement**
 - Engagement and feedback from end users via webinars, conference presentations, and consultations.
- **Peer Review**
- **Phase 3 – Model Application & Distribution**
 - Develop EPA website for SIACS
 - Configure a graphic user interface (GUI)
 - Develop guidance and messaging for various audiences
 - Publish SIACS model and associated content on EPA website
 - Establish a SIACS Community of Practice (tentative milestone)

A timeline of key project milestones according to the three project phases can be found in Table 3. Some project activities and their milestones occur concurrently. Additionally, some milestones may be subject to change as a result of unforeseen changes in personnel, funding, and management priorities.

Table 3. Project Timeline & Key Milestones by Project Phase

Phase 1: Modeling Needs and Requirements Analysis	
Determine customer and modeling needs; and project-specific requirements	Q4 2019
QAPP Development	Q4 2020
QAPP Review & Finalization	Q4 2021
Phase 2 – Model Development	
Model Design, Coding, & Optimization	Q3 2021
Model Testing	Q3 2021
Database Development	Q4 2022
Stakeholder Engagement	(ongoing)
Peer Review	Q1 2022
Phase 3 – Model Application	
Develop EPA Website for SIACS	Q3 2022
Configure a user interface	Q2 2022
Develop guidance and messaging	Q2 2022
Publish SIACS model on EPA website	Q4 2022

A5. Special Training/Certification

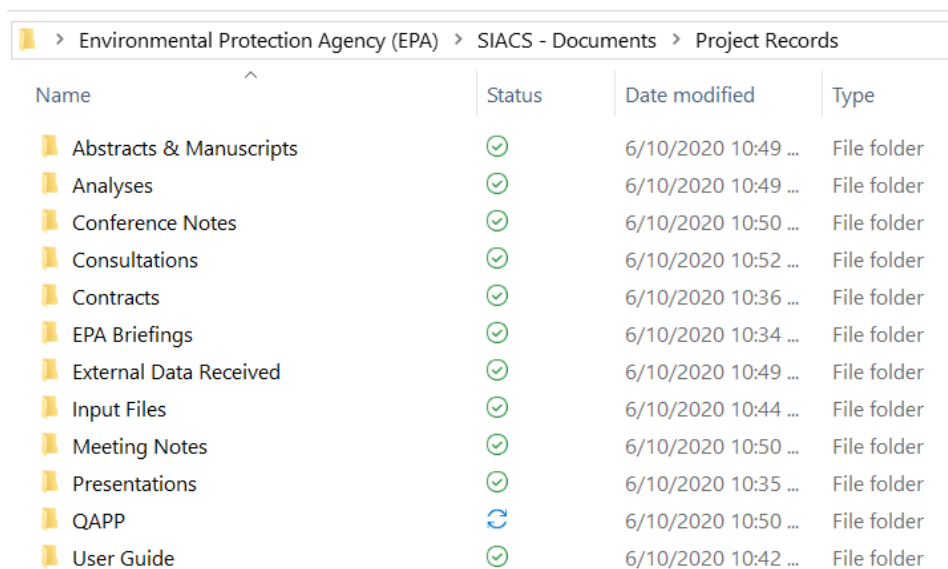
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No special training requirements or certification are needed to complete this project. General experience with common statistical analyses and familiarity with R programming language are desirable.

A6. Documents and Records

Upon approval, this QAPP will be distributed to individuals listed in Section A1. The QAPP will be maintained and accessible to project personnel on a shared project folder (see Section B10). During the project, any revision to this QAPP will be circulated to everyone on the current distribution list. The Technical Lead is responsible for ensuring all team members are distributed the current QAPP, including additional staff on their teams that enter the project between revisions.

Project records will be managed in accordance with EPA's National Records Management Program (NRMP). Records will be stored on the project share drive folder found at F:\ORIA\IED\SIACS and the project SharePoint folder, and email records will be saved according to EPA's capstone requirements and EZ email tool as appropriate. The SIACS model, related files, and previous versions will be maintained on the project share drive (F:). Project records, including analyses, user guides, related documentation, presentations, etc., will be stored on the SharePoint folder, and will be organized according to the following system, and updated as needed:



Environmental Protection Agency (EPA) > SIACS - Documents > Project Records			
Name	Status	Date modified	Type
Abstracts & Manuscripts	✓	6/10/2020 10:49 ...	File folder
Analyses	✓	6/10/2020 10:49 ...	File folder
Conference Notes	✓	6/10/2020 10:50 ...	File folder
Consultations	✓	6/10/2020 10:52 ...	File folder
Contracts	✓	6/10/2020 10:36 ...	File folder
EPA Briefings	✓	6/10/2020 10:34 ...	File folder
External Data Received	✓	6/10/2020 10:49 ...	File folder
Input Files	✓	6/10/2020 10:44 ...	File folder
Meeting Notes	✓	6/10/2020 10:50 ...	File folder
Presentations	✓	6/10/2020 10:35 ...	File folder
QAPP	↻	6/10/2020 10:50 ...	File folder
User Guide	✓	6/10/2020 10:42 ...	File folder

Figure 2. Example of Folder Organization

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Electronic and paper records, including email correspondence, will be managed according to EPA's NRMP¹. The following is a list of documentation to be referenced and records to be generated as a result of this process:

- Statements of work for contractor consultations regarding the SAICS model and performance
- Quality Assurance documentation submitted by contractors for modelling and application activities.
- Electronic records of secondary data collected from the EPA Office of Research and Development and other published and publicly available data.
- Compiled SIACS input data files. These files are the primary data source (input files) for the SIACS model.
- SIACS project files that capture model output, and Excel files summarizing the modeling results in these output files.
- Manuscript detailing the methods and results of the project.
- User guide for SIACS model.
- Presentations, briefings and conference materials.

A7. Quality Objectives and Criteria for Model Inputs/Outputs

A7.1 Overview

The project will use secondary (non-direct) data and will follow requirements outlined in EPA's *Quality Assurance Project Plan Requirements for Secondary Data Research Projects* (EPA, 1999), and EPA guidance for assessing the quality and credibility of collected data (EPA 2002; EPA 2003). Assessments of the quality and relevance of literature and data will follow the criteria for *Data Quality Assessment (DQA) of Data from Literature* outlined in Section A7.2. This criteria includes the soundness, applicability and utility, clarity and completeness, uncertainty and variability, comparability, and evaluation and review of the data being used. In conducting assessments of the quality and relevance of data, Data Quality Indicators (DQI) outlined in section A7.3 *Data Quality Indicators for Data* will be utilized. In applying these criteria, a graded approach will be used, ensuring that the degree to which quality requirements are applied is appropriate for the intended use and influence of the data.

Types of Reviewed Data

This project will collect secondary data on measured or modeled concentrations of indoor air pollutant species in outdoor and indoor air, atmospheric conditions and measurement conditions, and results of comparable existing modeling efforts. Secondary data collected for this project may be obtained from many sources, including literature and published research projects,

¹ EPA's National Records Management System: <https://intranet.epa.gov/records/tools/index.html>

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compilations from computerized databases and information systems, and computerized or mathematical models of environmental processes. These data are crucial for the development of the SIACS model and for its validation and verification. Secondary data collected for this project will be obtained primarily from peer-reviewed scientific literature. Peer-reviewed scientific literature generally reflects high levels of review and evaluation and where detailed methodology and supporting information is often available. The peer-review process places duties on the authors to ensure that the data are represented accurately. Also, the process places duties on the referees to examine a study's methodology and the precision of presented data. On that premise, these publications will be viewed generally as possessing the most reliable information, particularly if most of the criteria for *Data Quality Assessment (DQA) of Data from Literature* outlined in Section A7.2 are met. As available, publicly available datasets associated with peer-reviewed literature will be collected. The quality of these datasets will be scrutinized according to Data Quality Indicators discussed in section A7.3. Lastly, this project will use secondary data produced by EPA derived from EPA's Community Multi-Scale Air Quality (CMAQ) Model. Quality assurance documentation for this secondary data can be found in EPA *Quality Assurance Project Plan for the Development of Air Quality Simulation Models: Model Development Activities for Tropospheric Ozone, Particulate Matter, and Hazardous Air Pollutants (D-CED-000001-QP-1-0)*.

A graded approach will be used in this assessment, wherein the level of quality assurance applied to the information is commensurate with the intended use of the information, the degree of confidence necessary in that information, and the importance of the information to the project. Documentation of assessment activities will be maintained using the Documentation of Quality Control Form shown in Appendix A, Table A1. This form contains the data quality assessments (outlined in A7.2) and the data quality indicators (outlined in A7.3), both of which will be used for quality assurance purposes.

A7.2 Data Quality Assessment (DQAs) of Data from Literature

The principal criteria for evaluating literature will be their pedigree—i.e., whether the data appear in a reliable form, such as in a peer-reviewed publication – and relevance to the task at hand. The assessment will also consider how far removed the data are from the original source (i.e., is it a primary source or a secondary source drawing data from the primary source). Literature will be evaluated using the five assessment factors described by EPA's Science and Technology Policy Council in *A Summary of General Assessment Factors for Evaluating the Quality of Scientific and Technical Information* (EPA 2003). Reviewers will use this criterion when evaluating literature identified for possible inclusion. The decision tree presented in Figure 3 incorporates the quality assessment factors presented in Table 4 and will be used in the data and literature evaluation.

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Table 4. Assessment Factors and Assessment Criteria for Data from Literature

Assessment Factor	Summary Assessment Criteria for this Project
Applicability and Utility	The publication provides relevant information for the task at hand. Types of questions considered in this assessment factor will be the following: Does the publication present information, such as mathematical relationships of indoor conditions and air pollutant species, applicable to those needed for this project? Specifically, are the measurements or mathematical relationships reflective of a residential environment?
Soundness	The publication and any conclusions reported rely on sound scientific theory and approaches, and the conclusions are consistent with the data presented. Types of questions considered in this assessment factor will be the following: Are the methodology used for data measurement appropriate? How do the results compare with existing scientific theory, principles, and practices?
Clarity and Completeness	Sufficient documentation is provided with the information/data being considered to assess underlying data, key assumptions, procedures, and model parameters, as applicable, as well as information identifying any potential conflicts of interest about sponsorship and author affiliations. Types of questions considered in this assessment factor will be the following: Are presented mathematical relationships supported by established theory, principles and/or practice. Are simulated data and derived parameters supported by clear and reproducible methodology?
Uncertainty and Variability	Sufficient documentation is provided with the information/data being considered to assess uncertainties, variability, sources of error and/or bias, and properly reflects them in any conclusions drawn.
Evaluation and Review	The information/data has been independently technically reviewed, peer reviewed, or cited and/or used by a project that has been so reviewed or has documentation of independent verification or validation of the data study method and results reported.
Comparability	Can the data be compared across datasets? Are the results presented in a manner that is consistent with the objectives of the study?

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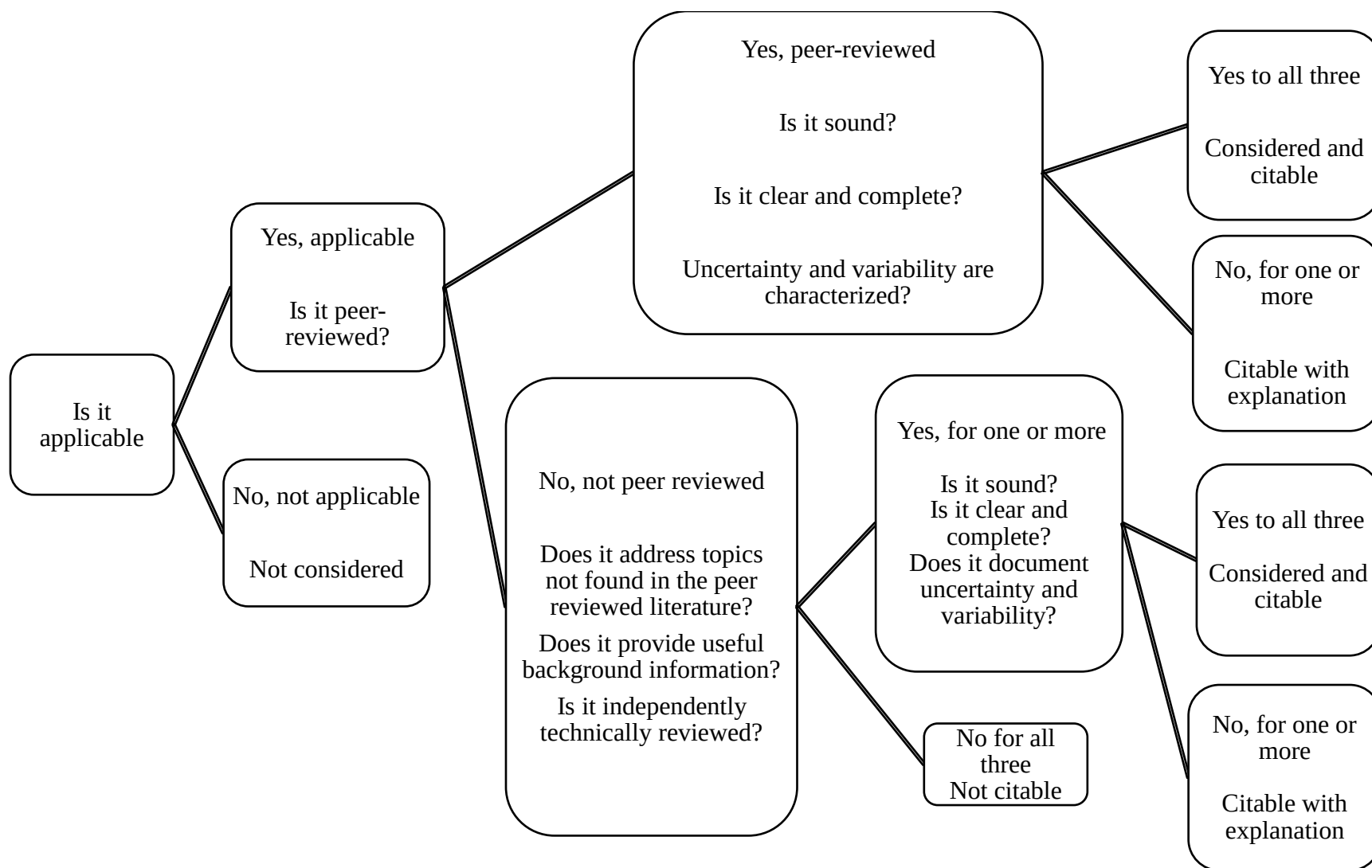


Figure 3. Process for Literature Evaluation

A.7.3 Data Quality Indicators (DQIs) for Information/Data

The DQI will assist in determining whether the data are suitable for the intended use in this analysis. That is, is the data useful in understanding temporally-resolved concentrations of various air pollution species in residential settings? The DQI will follow that provided in *Guidance for Quality Assurance Project Plans QA/G-5* (EPA 2002). In implementing the graded approach, each of the following six data quality indicators will be considered as applicable to the type of data being assessed.

1. **Precision:** Information on indoor species are expected to have been obtained from direct measurement.
2. **Bias:** The potentially biased information can be appropriately used in ascertaining bounding analyses and may be useful to identify limits to possible parameter values. If this is the case, the rationale as to where the bias may lie and why it is being used is provided.
3. **Representativeness:** Data on indoor and outdoor air pollution species in residential settings will be utilized. Representativeness is a qualitative term that expresses the degree to which information/data being considered accurately represents the population and/or parameter we are trying to address
4. **Completeness:** The review team will look for indications that information sources are well documented and that the study has been independently reviewed and approved.
5. **Comparability:** Comparability is a qualitative term that expresses the measure of confidence that the information/data from different reports on the same or similar subjects/concepts can be compared and combined for evaluation and development of decision(s) related to the project tasks at hand.
6. **Sensitivity:** The sensitivity of project results to specific data will be an important consideration in the use of those data in the study. The precision and accuracy of data will be carefully evaluated in a graded QA approach and uncertainty bounds may be established in the analysis for sensitive data of questionable accuracy.

B. MODEL DEVELOPMENT

B1. Model Description

This project consists of model development tasks such as model architecture design, coding, testing, documenting, deployment, version control, and issue tracking. The project will use secondary (non-direct) data from many sources and may generate large volumes of modeled data during implementation. No sampling or collection of primary data will occur.

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The SIACS model is a deterministic, dynamic model of gas and particle concentrations in indoor air. It assumes an idealized building consists of a single, well-mixed zone. The five processes included in the model are: 1) infiltration from the outdoors, 2) exfiltration to the outdoors, 3) indoor photochemical reactions, 4) indoor emissions, and 5) irreversible indoor deposition on surfaces. Figure 4 displays the input variables needed in the SIACS model to develop the time-resolved concentrations of indoor air species.

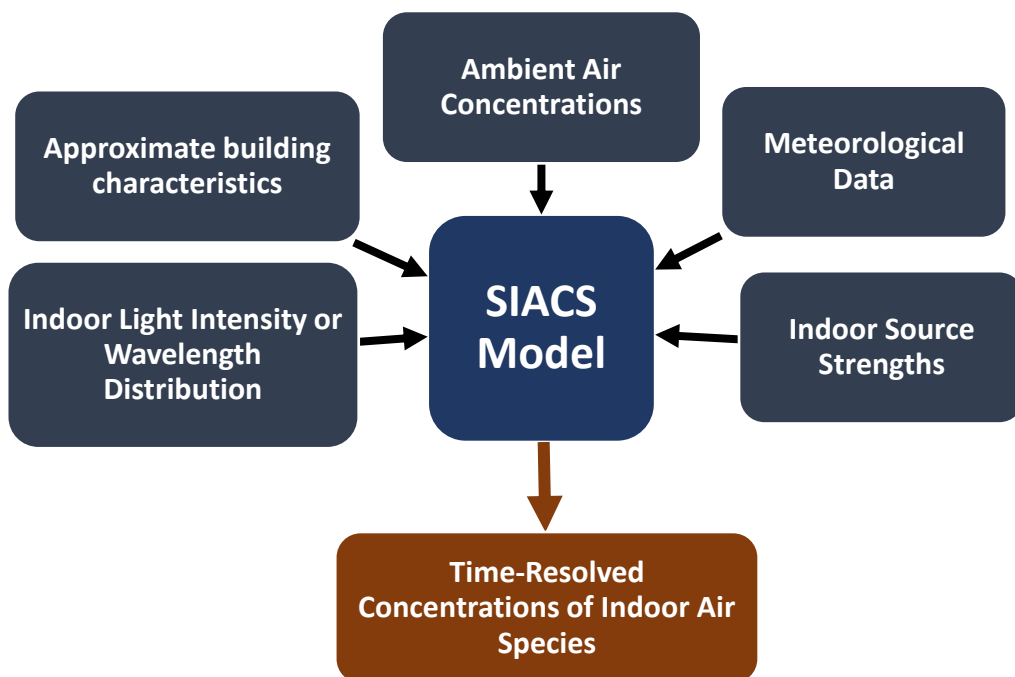


Figure 4. Five Components of the SIACS Model

B2. Model Parameters & Theoretical Approach

Indoor concentrations of species are modeled by the mass conservation equation (Koutrakis et al. 1992):

$$\frac{dC_{i,x}}{dt} = PaC_{o,x} - aC_{i,x} + \sum_{n,m} \pm k_n C_{i,n} C_{i,m} + \frac{S_x}{V} - v_x \frac{A}{V} C_{i,x} \quad (\text{eq. 1})$$

where:

$C_{i,x}$ indoor concentration of species x

$C_{o,x}$ outdoor concentration of species x

P penetration factor (across building envelope)

a air exchange rate (between indoor and ambient air)

k_n pseudo-first order reaction rate constant between species indoors

S_x emission rate of species x from indoor sources

v_x indoor deposition velocity for species x

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- A indoor surface area
V indoor volume

The infiltration contribution to the air exchange rate is calculated using the extended Lawrence Berkeley National Laboratory parameterization (Breen et al 2010; Breen et al. 2014; ASHRAE 2017), based on temperature differential and wind speed to calculate infiltration and natural ventilation, as well as any mechanical ventilation rates (assumed as known inputs). The total ventilation rate is calculated as

$$Q_{tot} = Q_{bal} + \sqrt{Q_{unbal}^2 + Q_{nat}^2 + Q_{inf}^2} \quad (\text{eq. 2})$$

$$Q_{inf} = A_{inf} \sqrt{k_s |T_i - T_o| + k_w U^2} \quad (\text{eq. 3})$$

$$Q_{nat} = \sqrt{Q_{nat,wind}^2 + Q_{nat,stack}^2} \quad (\text{eq. 4})$$

where:

- Q_{tot} total ventilation rate
 Q_{bal} balanced mechanical ventilation rate
 Q_{unbal} unbalanced mechanical ventilation rate
 Q_{nat} natural ventilation rate (through windows and doors)
 Q_{inf} infiltration (leakage) rate (through cracks in the building envelope)
 A_{inf} Area across the building envelop
 K_s Stack Coefficient
 T_i Temperature indoors
 T_o Temperature outdoors
 K_w Wind Coefficient
 U Wind Speed
 $Q_{nat,wind}$ airflow from wind effect
 $Q_{nat,stack}$ airflow from stack effect

The photochemical reaction model is based on the condensed SAPRC-99 mechanism, originally designed for volatile organic compound (VOC) reactivity assessment in regional models (Carter, 2000). The original mechanism was developed to assist in identifying the secondary pollutant formation potential of different VOCs, and this focus on secondary reaction byproducts remains important for health applications indoors. Additional model development will address the significance of changing the modeling environment and the assumptions implemented when applying the mechanism indoors in order to adapt the chemical mechanism accordingly.

An additional component will be developed to allow for the modeling of indoor ultrafine particles (UFP), particulate matter with diameters that are $\leq 2.5 \mu\text{m}$ ($\text{PM}_{2.5}$) and particulate matter with diameters that are $\leq 10 \mu\text{m}$ (PM_{10}) based on ambient and indoor sources and settling on surfaces. The component will allow the model to account for sources or reactions happening

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outside of the particles phase –including SOA formation and surface chemistry. The current version of the model assumes all reactions take place in the gas phase and no heterogeneous interactions, including surface absorption and desorption, are represented in SIACS. Lastly, a component for modeling the interaction of light will be developed to improve the model's ability to account for interactions between light and various species.

B3. Non-direct Measurements (Input Data Requirements)

The project will use secondary (non-direct) data from many sources and may generate large volumes of modeled data during implementation. Most parameters (Tables 5 and 6) consist of default values provided by IED, while others are provided as inputs from users who will access the model code or web application. Explanation of default values will be provided to users in the SIACS User Guide. Other data sources, utilized by the end user and not provided by EPA, will be subject to the quality guidelines of the end user and their project objectives. It is anticipated that all secondary data will be acquired in electronic format. Acquisition of data will be documented in electronic repositories. Metadata concerning each of the model's inputs and outputs will be stored on the project SharePoint and will also be made available with the final public release of the model. Units are based upon the specification of the individual model components.

Within the limitations of the model, users can modify most variables represented in equation 1. Some input variables must be assigned a fixed value for the entire modeled time period, others can be modeled as dynamically changing over time. Dynamic variables represent evolving characteristics of the indoor and ambient atmospheres. The fixed variables are conceptualized as representing relatively unchanging characteristics of the indoor space (building) being modeled. The variables governing the photochemical mechanism are thought as fixed (and optimized for the application) and cannot be changed without modifying the code for the R scripts. The following tables present the variables that can currently be changed without changing the R script for the model.

SIACS runs for a specified length of simulated time (in minutes) and will generate results with a specified resolution. The solver in the model, however, calculates time steps as needed for solution convergence, irrespective of this resolution.

A key feature of SIACS is the ability to work with limited data. All time-dependent input variables can take a specified value as frequently or as sparsely as available data allows, but as a minimum must be defined at the start and end times of the simulation. SIACS interpolates the available data for every time dependent variable as needed, using the monotonic Fritsch and Carlson spline method. Since this method does not guarantee positive values, a modification has been developed to ensure non-negativity and continuous first-order derivatives..

The initial indoor concentrations can be provided based on known data, or calculated to match equilibrium conditions at the start of the simulations. In the latter case, it is recommended that users allow the model to run under constant conditions matching the desired initial state. This model spin-up time varies based on the conditions modeled, but it is longer with lower air-exchange rates.

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Table 5. Input Variables with Time-Dependent Values

Condition Variables		Units
<i>Ambient</i>	T_o , Outdoor temperature	K
	Wind speed	m/s
	$C_{o,i}$ Outdoor concentrations of species	ppmv; $\mu\text{g}/\text{m}^3$ (for particles)
	Barometric pressure	Pa
<i>Indoor</i>	T_i Indoor temperature	K
	<i>SunFactor</i> (indoor fraction of zenith sun-light (summer conditions and directly overhead) integrated photon flux, 300-760 nm),	[0,1]
	S_x , Emission rate of species	g/min
	Ventilation rate (over and above the infiltration rate)	m^3/hr
	Relative humidity	[0,1]

Table 6. Descriptions of Input Variables Related to Building Characteristics

Variable	Description	Units
Floor surface	Floor surface area of space modeled	m^2
Room height		m
Area to volume ratio		m^{-1}
Infiltration surface area	Effective leakage area for infiltration	m^2
P_i	Gas Penetration factor (for species i)	Unit-less
k_w	Stack coefficient (Breen, 2010)	$(\text{L}/\text{s})^2/(\text{cm}^4\text{K})$
K_s	Wind coefficient (Breen, 2010)	$(\text{L}/\text{s})^2/(\text{cm}^4(\text{m}/\text{s})^2)$
V_i	Deposition velocity for species i	m/min

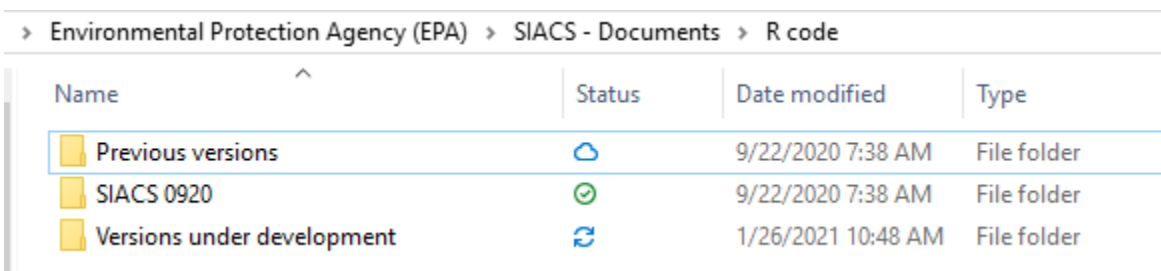
An aim of this project will be to collect input parameters from the literature and directly from stakeholders and partners, and to store these data in a database. Any collected literature will be evaluated according to Quality Assessment Factors described in Section A.7.2. We will construct this database in Excel and may eventually transition the database to a more user-friendly software. The Excel database will provide detail on the parameter name, value and citation.

B4. Model Coding and Verification

1. **Code development.** During the code development records of all the assumptions, parameter values and their sources, model inputs and outputs and their formatting requirements will be documented. Once the code development is complete, we will prepare a code documentation that will include model specifications, model description, flow charts, description of routines, source listing, and error messages.

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The image below displays the location where the previous versions, current version and versions under development of the modeling code will be stored on the project share drive (F:). As mentioned in Section A6, all official records will also be saved on EZ Records. All new versions will be released with updated documentation, which will summarize the model changes made to the new version. The previous version and documentation will be preserved and stored in a specified project folder named "Previous versions." The new version will be stored in the same project folder and named "SIACS" followed by its release date, month <MM> and year <YY>. For example, the current version is named "SIACS 0421" because it was released (internally to the development team) in April 2021. This process is repeated as new versions become available. The Principal Investigator (PI) is the gatekeeper of the current model code and any modifications and additions to the code need to be verified and approved by that individual. Model code that is under development is stored in the folder "Versions under development." With regards to version control, all changes in the code are highlighted using the comment (#) symbol and tracking of modifications is done manually (generally in a word document) by the code developer. Once the model is publicly available, the code may be moved to the EPA GitHub where subsequent version control and tracking automatically may be accomplished.



Environmental Protection Agency (EPA) > SIACS - Documents > R code			
Name	Status	Date modified	Type
Previous versions	Cloud icon	9/22/2020 7:38 AM	File folder
SIACS 0920	Green checkmark icon	9/22/2020 7:38 AM	File folder
Versions under development	Refresh icon	1/26/2021 10:48 AM	File folder

2. **Computer requirements for hardware and software.** The selected programming language for this project is R programming language. The source code will be made available to the public at the end. The code will be compiled with R Studio and the resulting model will be run on any PC or Macintosh computer using the R or R Studio software.
3. **Code verification.** The code will be documented with statements describing the purpose and details of a code/programming line or group of programming lines, including functions and subroutines. To verify the developed code, each new function, subroutine or calculation performed by the code will be compared to results of test-case calculations performed outside SIACS. These, in turn, will be reviewed by at least one more person in the team before the comparison with code results is performed. In case of a mismatch, the code will be inspected line by line for potential mathematical errors and/or bugs. If the code is not functioning in a test trial, a R Studio debugger will be launched to identify potential errors and attempts will be made to diagnose the source and correct the error.

B5. Sensitivity and Uncertainty Analysis

In the context of this modeling effort, sensitivity analysis is intended to evaluate the degree each input uncertainty contributes to the uncertainty of the model output. It is expected that relationships between many variables will be highly non-linear, and consequently the sensitivity to changes in inputs needs to be assessed at multiple possible values of each variable. The sheer number of variables in SIACS precludes a full sensitivity analysis where the impact of each variable is assessed individually.

Modeling uncertainty results from approximating the physical environment with a mathematical model. Although the assumptions involved in mass balance, in the photochemical mechanism, the infiltration model, and other components have been separately evaluated and found to be adequate for most of their intended purposes, their combination in a single model has not yet been evaluated against empirical data. Such comparisons, when available, will be included in this user guide to guide users in choosing appropriate applications of SIACS.

Measurement uncertainty results from imperfect knowledge of the true value of measured variables. To the extent that uncertainty exists within each of the input variables to SIACS it will also exist in the output concentrations as well. The quantification of measurement uncertainty, through error propagation or other method, in the model is under development and is planned as a feature of the first public version of SIACS. This uncertainty characterization is important in setting apart SIACS as a useful tool in making decisions about indoor air quality.

Monte Carlo simulations will also be performed in this project. Monte Carlo simulation performs risk analysis by building models of possible results by substituting a range of values—a probability distribution—for any factor that has inherent uncertainty. It then calculates results over and over, each time using a different set of random values from the probability functions. We will populate input parameter sets by random generation of the input parameters from their respective a-priori distributions and run the model with each parameter set. We will rely on expert judgment and literature for parameter distributions. If no information is available, uniform distribution will be assumed (uninformed priors).

B6. Model Calibration

This modeling effort will not require the calibration of analytical instruments for the purposes of generating analytical data. However, this effort will require model calibration, or the estimation of parameters involved in the formulation of various empirical relationships. During the development phase, some model parameters may be calibrated using existing observational data from the scientific literature and associated data, all of which are supplied and maintained by external sources and groups (such as [HOMEChem](#)). Quality assurance documentation associated with data obtained from these external sources and groups will be provided and reviewed according to DQA and DQI outlined in section A7. Individual researchers on this project are responsible for vetting any acquired data sets as to their general quality by scrutinizing such aspects as the data uncertainty, completeness, and appropriateness (and other applicable aspects

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as those outlined in section A7) for the modeling task at hand. This quality check will be recorded in a table format on the Documentation of Quality Control Form as mentioned in section A7.

Calibration of parameters via model-fit comparison of predicted vs. observational data

Model simulations will attempt to reproduce the observed data. For a select set of parameters or modeling options, optimization will be used to minimize residuals of indoor concentrations. Some examples of modeling parameters or options candidates for optimization include deposition velocities, mechanism of heterogeneous HONO production, and penetration factors. The selection of parameters to calibrate and the associated values implemented into the model will be at the discretion of the group team and the Principle Investigator (PI) as well as the availability of the parameters in existing studies. As mentioned above, the PI is the gatekeeper of the model code, which also includes accompanying parameters.

Inputs to the model calibration process can highly influence the quality of information generated by the model. Therefore, the calibration process may need to be iterative in nature, repeated whenever some key aspect of the environment changes. Each iteration would utilize data that accurately portray the changing environment and, therefore, would provide further necessary refinements to the model.

For convenience and documentation purposes, a calibration template was created that contains entries for the modeling options, parameters and their ranges as well as associated model results and residuals. Documentation of calibration activities will be maintained using the Calibration Template shown in Appendix A, Table A2. When conducting calibrations, the researcher will use the calibration template to ensure inputs and outputs of the model are recorded. Statistical methods, such as regression analyses and goodness-of-fit methods will be used when performing model calibrations. The details of the model calibration procedure, including any statistical analyses that are involved, and technical notes will also be documented on the calibration template.

The model will be calibrated by comparing the predictions (output) for a given set of assumed conditions to observed data for the same conditions. This comparison will allow the modeler to evaluate whether the model and its parameter reasonably represent the environment of interest (i.e., indoor environment). The acceptance criteria will depend on the Relative Root Mean Square Error (RMSE) where a value of zero indicate a perfect fit of the modelling results to the observed data. Lower values, generally, indicate a better model fit than higher values. The acceptance criteria for the use of a specific parameter is 1) there are no other options found in the literature or 2) when compared to other values obtained from literature, the RMSE associated with this specific parameter is lower than the other RMSEs.

The calculation for the RMSE is shown on the calibration template, and a selected set of species will be used as inputs (instead of having all 78 species used as inputs). For this smaller set,

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species will be selected based on how impactful they are to indoor chemistry and how significant they are to human health. The selected set of species will be documented once all of them are identified.

B7. Model Verification (Model Comparison)

The model will be verified concurrently using existing analytical tools for indoor chemistry simulation from the peer reviewed literature. Specifically, we will compare simulated concentrations of key species (NO, NO₂, HONO, O₃, HCHO and OH) using SIACS to previously published values in the literature. For example, we will compare SIACS output with values from the Indoor Chemistry and Exposure Model (ICEM) by Sarwar et al. (2002), the detailed chemical model by Carslaw (2007) and the INCA-indoor model by Mendez and coworkers (2015), and the mathematical model by Nazaroff and Cass (1986). Analyses will focus on perturbing the base case scenario to assess SIACS performance in modeling changes in concentrations, and these results will be compared to the previously published models. Process of performing these model comparisons and their results will be summarized in a white paper(s), conference papers and/or published manuscript(s) in peer-reviewed journals.

B8. Model Evaluation & Validation

The SIACS model, SIACS User Guide, and other associated products will be evaluated in three phases: (1) Simultaneous review by EPA reviewers and non-EPA Federal partners with expertise on indoor air chemistry; (2) review of SIACS model and User Guide by a small group of external modelers with indoor air chemistry and modeling expertise; (3) review of the model and key scientific results and findings by a scientific journal.

These three stages of evaluation will be conducted to evaluate, adjust, and refine the model based on performance (Figure 4). These three phases of testing will include testing all aspects of the SIACS model, including quality assurance testing for functionality, Section 508 compliance, spelling and grammatical errors, and other technical specifications.

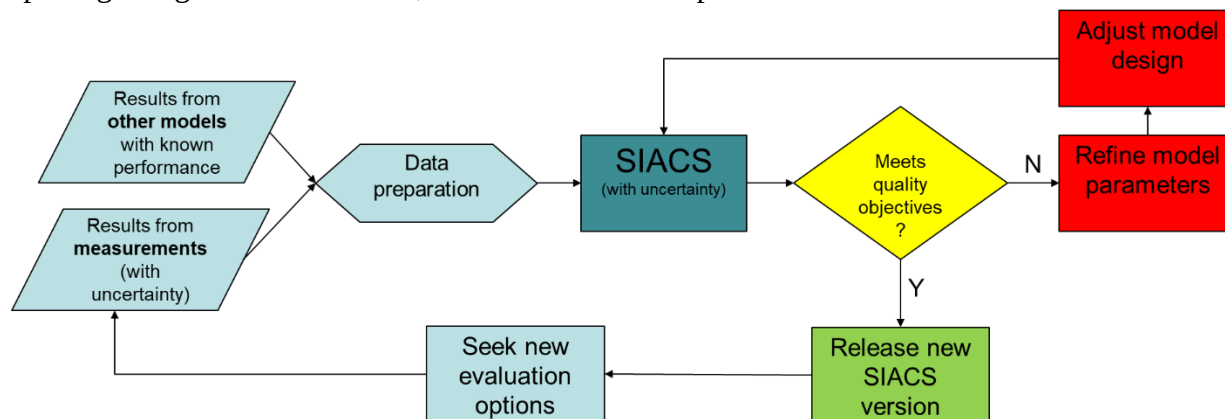


Figure 4. Model Evaluation Process

B8.1 Model Evaluation Plan

- **Phase 1. EPA and Federal Subject Matter Expert Review**

In an initial phase (Phase 1), the model will be independently and concurrently reviewed by: EPA subject matter experts and external Federal subject matter experts. All subject matter experts will be selected based on their expertise in indoor air chemistry, atmospheric chemistry, and modeling, and will be requested to review and apply the model to their data. A request for external Federal reviewers may potentially include scientific expertise from the Consumer Product Safety Commission, Department of Energy, National Institute for Occupational Safety and Health, Occupational Safety and Health Administration, Lawrence Berkeley National Laboratory, and the National Institute of Standards and Technology. In preparation for these concurrent reviews, the project team will develop validation criteria for the review (separately of this QAPP). Reviewer feedback will be collected, revisions to the model will be implemented, and responses will subsequently be provided to reviewers on how their feedback was addressed and how associated improvements to the model were made. Once the validation criteria (to be developed separately) and performance of the model based on this feedback and subsequent revisions is satisfactorily met, a second phase of evaluation will be conducted.

- **Phase 2. Model Usability Review**

In the second phase (Phase 2), the model will be disseminated to a small group of external and EPA modelers, identified among existing partners from academic and government institutions who have expertise in indoor air chemistry, for beta testing of the model and GUI, to test its usability and functionality.

- **Phase 3. Scientific Peer Review**

Finally, in the third phase (Phase 3), application of the model and its results will be submitted for review by a scientific journal. It is anticipated that in this third stage, the results of this study will undergo internal technical review for presentations and will also undergo external scientific peer-review before results are published in either an EPA report or peer-reviewed journal. Additionally, the manuscripts will also undergo internal technical reviews by individuals selected for their expertise related to this field prior to publication clearance. These reviews are documented using associated internal routing and clearance methods.

B8.2 Acceptance Criteria and Documentation

In all evaluation and review processes, specific evaluation and acceptance criteria, reviewer

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comments and EPA's response to comments (which will address steps EPA has taken to address reviewer feedback) will be documented. Acceptance criteria (to be prepared separately in preparation for the Phase 1 and Phase 2 reviews) will be reviewed and agreed upon by the project team as part of the model review process. The following provides an overview of the prospective process by which the product will be validated to ensure it meets the acceptance criteria.

1. The model version and associated files are made available in a testing environment.
2. Tester(s) will test the functionality of the model based on agreed upon requirements (to be prepared separately in preparation for the Phase 1 and Phase 2 reviews). For example, testing for correctness of functionality can be based on the model generating outputs within $\pm 1\%$ of each other for the same set of input data, when run without stochastic variables (e.g. measurement uncertainty).
3. The technical lead will report as bugs any acceptance criteria that are not met and will assign them to a project team member for adjudication (refinement/adjustment of model/files).
4. The technical lead will prioritize bugs to be fixed ahead of new features.
5. Developers fix all reported issues.

More information on model testing and validation can be found in **Section D1. Model testing strategy**. Records will be managed according to processes outlined in **A6. Documents and Records** for more information.

B9. Data Management

All project files (statements of work, QAPPs, data files, etc.) are stored on a private EPA SharePoint drive, in a folder labeled "SIACS". Files are backed up on a regular basis (weekly & monthly). Access is restricted to those on the project team. The SIACS model, related files, and previous versions will be maintained on the project share drive (F:). As mentioned in Section A6, all official records will also be saved on EZ Records.

This project will acquire data from secondary sources, manipulate secondary data to meet user requirements, generate data from model output, and create summary and processed files for further statistical and analytical treatment. It is anticipated that the project will acquire or generate little or no hardcopy data. The project will maintain a data management system that protects integrity of the data received and generated throughout the project. This includes file management systems, version control, archiving procedures, and quality assurance activities. See **A6. Documents and Records** for more information.

C. ASSESSMENTS AND OVERSIGHT

C1. Assessments and Response Actions

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QA Audits/Reviews. The ORIA QAM, IED QAC, as well as other designated QA personnel for this project may choose to conduct a Technical Systems Audit (TSA) during execution of the research project to ensure this QAPP is being implemented as intended. Alternatively, QA staff, or a designee, may perform an audit of data quality on either all or a portion of the generated data if requested by the primary client (IED/ORIA Management). QA evaluations may include interviews with development team members, and a review of research project work and quality records. Any findings, research best practices, or improvements will be compiled in a report and filed with other QA documents. Any corrective actions and responses will be documented and filed with the Assessment Report and stored by the project lead. The project team leader will have responsibility for monitoring project activities and identifying or confirming any quality problems. Any problems will be brought to the attention of the IED Management Team, IED QAC, and the ORIA QA Manager, who will initiate corrective actions, document the nature of the problem, and ensure that the recommended corrective action is carried out.

Additionally, any associated scientific and technical products cleared through ORIA/IED will also undergo a quality assurance review prior to clearance to ensure that the developed product was within the scope of the appropriate QAPP.

C2. Reports to Management

Weekly meetings with the project and task lead, as well as other team members, will be held to verbally report on the status of the project, results of performance evaluations and systems assessments, results of data quality and modeling evaluations, and any significant quality assurance problems and recommended solutions. Documentation of research progress is kept in SharePoint project notes.

Periodic updates/progress reports will be given to management as needed to discuss the progress of the project. Recommended management actions will be taken as appropriate to adjust the research. Any identified errors, deficiencies, and anomalies will be documented and reported to the team. If needed, an error analysis will be performed, and the model processes will be reviewed and/or modified. Any identified deviations from the approved QAPP will be documented and reported to the task lead, management, QAC and QAM. Electronic quality control information, technical assessments, and documentation will be made available to interested parties.

D. MODEL VALIDATION AND USABILITY

D1. Model testing strategy

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The model testing strategy will be developed separately of this QAPP and in preparation for the Phase 1 and Phase 2 of the model evaluation, to ensure the final model meets the design objectives (see section **B8. Model Evaluation & Validation**). As such, the testing protocols will focus on whether or not the model:

- Provides reproduceable and sound model output
- Is user friendly
- Does not fail in unpredictable manners when improper data are input
- Satisfies specified requirements as outlined in this QAPP

To meet these objectives, we will apply a mixed approach for quality assurance, consisting of the following components:

- Make beta versions of the program available to interested parties to evaluate the model's ease of use and documentation. This early implementation of model versions with customers will create early feedback.
- This feedback will be considered for a revision of the model if necessary and it will be important for the continued work.
- Consultation with programming subject matter experts.
- Implementation of consultation feedback for optimization of model performance.

D2. Procedure for checking the correctness of outputs

With respect to coding development, model outputs will be assumed to be without error except for known bugs in R programming software previously documented internally or posted online. SIACS model outputs will be verified through running and checking results associated with the modified code. See section **B4. Model Coding and Verification** for more information.

Model verification and the checking and vetting of model output will be conducted according to the processes outlined in sections **B6. Model Calibration** and **B7. Model Verification (Model Comparison)**.

D3. Quality assurance of code development.

Model code are checked by members of the model development staff. Checks include performing testing of individual scripts and the integrated model. Quality objectives and criteria of the developed code include accuracy, in-line documentation, code readability, adherence to coding standards, model scripts, archival and version control. A model bug-reporting system is in place through the R system to report bugs or code errors to be investigated. The development team will explicitly declare all variables used, utilize R language constructs and conventions, and include comment lines explaining algorithms and descriptive variable names with units listed in the code or in supporting documentation. The SIACS User guide will provide additional supporting information to assist the user with understanding the model code, input and output variables, and overall construct.

D4. Conformance to requirements

Conformance to requirements will be evaluated to determine whether the model can meet the prescribed functional requirements. Limitations of the model will be noted within the model itself and as part of the user instructions.

E. REFERENCES

- ASHRAE, 2017. Handbook-Fundamentals, Atlanta: American Society of Heating, Refrigerating, and Air-Conditioning Engineers, Inc.
- Breen, Michael S., Bradley D. Schultz, Michael D. Sohn, Thomas Long, John Langstaff, Ronald Williams, Kristin Isaacs, Qing Yu Meng, Casson Stallings, and Luther Smith. "A review of air exchange rate models for air pollution exposure assessments." *Journal of Exposure Science and Environmental Epidemiology* 24, no. 6 (2014): 555
- Breen, Michael S., Miyuki Breen, Ronald W. Williams, and Bradley D. Schultz. "Predicting residential air exchange rates from questionnaires and meteorology: model evaluation in central North Carolina." *Environmental science & technology* 44, no. 24 (2010): 9349-9356.
- Carslaw N. 2007. A new detailed chemical model for indoor air pollution. *Atmospheric Environment*. 41, 1164–1179.
- Carter, W. P. 2000. Documentation of the SAPRC-99 chemical mechanism for VOC reactivity assessment. Contract, 92(329), 95-308.
- Koutrakis P, Briggs SLK, Leaderer BP . Source apportionment of indoor aerosols in Suffolk and Onondaga counties, New York. *Environ Sci Technol* 1992; 26: 521–527
- Mendez M, Blond N, Blondeau P, Schoemaeker C, Hauglustaine DA. Assessment of the impact of oxidation processes on indoor air pollution using the new time-resolved INCA-Indoor model
- Nazaroff WW and Cass GR. 1986. Mathematical Modeling of Chemically Reactive Pollutants in Indoor Air. *Environmental Science and Technology*. 20:9, 924-934.
- U.S. Environmental Protection Agency. 1989. Report to Congress on indoor air quality: Volume 2. EPA/400/1-89/001C. Washington, DC.
- U.S. Environmental Protection Agency 2002a. Guidelines for Ensuring and Maximizing the Quality, Objectivity, Utility, and Integrity of Information Disseminated by the Environmental Protection Agency. EPA/260R-02-008 (October 2002 with Addenda in 2004 and 2005).U.S. Environmental Protection Agency.
- U.S. Environmental Protection Agency 2002b. Guidance for Quality Assurance Project Plans for Modeling EPA QA/G-5M

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U.S. Environmental Protection Agency 2003. A Summary of General Assessment Factors for Evaluating the Quality of Scientific and Technical Information (EPA 100/B-03/001) June 2003 and addendum, U.S. Environmental Protection Agency.

F. LIST OF SECTIONS NOT APPLICABLE TO THIS QAPP

Sections B.1-B.6 and B.8 of the [QAPP GUIDANCE (i.e., Sampling, Measurement, etc.)] are not applicable to the scope of this project or QAPP.

- B1. Sampling Process Design
- B2. Sampling Methods
- B3. Sample Handling and Custody
- B4. Analytical Methods
- B5. Quality Control
- B6. Instrument/Equipment Testing, Inspection, and Maintenance
- B8. Inspection/Acceptance for Supplies and Consumables

Appendix A

QC Documentation Table

Table A1. Data Quality Assessment Factors (DQAs) and Data Quality Indicators (DQIs) review of existing information/data

Data Source Citation		Data Quality Assessment Factors						Data Quality Indicators							Explanations
		Comparability	Evaluation and Review	Uncertainty and Variability	Clarity and Completeness	Soundness	Applicability and Utility	Sensitivity	Comparability	Completeness	Representativeness	Bias	Precision		
No.	Information data source citation	X	X	X	X	X	X	X	X	X	X	X	X		
							</								

Calibration Template

Table A2. Calibration template						
What is the source of the parameter being evaluated?	What is the source of the observation data?	What parameters are being evaluated?	Where are the simulation conditions? What are the inputs?	What outputs are of interest?	Relative root mean square error (RMSE)	Notes

$$Relative\ Difference = 100 \times \frac{SIACS_{avg} - observation_{avg}}{observation_{avg}}$$

The relative difference is per-species, and it is the average over the time period.

$$Relative\ RMSE = \sqrt{\sum (Relative\ Difference)^2}$$